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Infrastructure as a Service

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Contract Opportunity

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Related Notice

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DEPT OF DEFENSE

General Information

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Original Set Aside:

Product Service Code: 7D20 - IT AND TELECOM - SERVICE DELIVERY MANAGEMENT (HARDWARE AND PERPETUAL LICENSE SOFTWARE)

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- 541519 - Other Computer Related Services

Place of Performance:

Aberdeen Proving Ground , MD 21005
USA

Description

Program Executive Officer for Command, Control, Communications-Network (PEO C3N) Project Manager Mission Command (PM MC), Product Manager Tactical Mission Command (PdM TMC) is requesting interested sources to provide information in the form of a white paper which describes an approach to providing Infrastructure as a Service (IaaS), as outlined in this Request for Information (RFI). Information shall be submitted electronically and shall not exceed ten (10) pages for all items associated with this RFI.

1. Request for Information

PdM TMC is soliciting industry input on new and innovative approaches for contracting for, provisioning, maintaining, and updating the computing resources/environments needed to support unit Mission Command in all possible operational environments, including home station operations, exercises, experiments, training center rotations, and overseas operational deployments in denied, degraded, intermittent, and low-bandwidth (DDIL) environments. Additionally, capacity must be available for testing, materiel release approval (IAW AR 770-3), continuity of operations, overcoming hardware failures, and support of multiple information domains during coalition operations.

Industry solutions may either supplement or replace our current way of doing business. Solutions that require alteration to Army Regulations or Policies will be considered but must identify specific changes. Solutions that require alteration to DoD Regulations, Federal Regulations or Statutory law will not be considered. The ideal IaaS offering will provide reliable, on-demand access to virtualized computing resources, to facilitate cost-effective and efficient operations. Solutions should facilitate seamless unit transitions from home station to deployed environment/exercise and return. We aim to enhance our ability to deploy, manage, and scale these mission-critical applications and services rapidly while maintaining a secure and compliant environment. The solution should support various workloads, including development, testing, large scale combat operations, and disaster recovery/ Continuity of Operations (COOP).

PdM TMC is requesting interested sources to provide information in the form of a white paper which demonstrates the capability to meet the requirements described within this RFI. Information shall be submitted electronically and shall not exceed ten (10) pages for all items associated with the RFI response.

This is a RFI only. This RFI shall not be considered an Invitation for Bid (IFB), Request for Task Execution Plan (TEP), Request for Quotation (RFQ) or a Request for Proposal (RFP). No solicitation document exists, and a formal solicitation will not be issued by the Government as a result of the responses to this RFI. The information provided will be used by the Government for the purpose of conducting market research. There is no obligation on the part of the Government to acquire any products or services described in this RFI or its responses. Responses to this RFI will be treated only as information for the Government to consider. There is no payment for direct or indirect costs that are incurred in responding to this RFI. This request does not constitute a solicitation for proposals or the authority to enter negotiations to award a contract. No funds have been authorized, appropriated or received for this effort. Interested parties are responsible for adequately marking proprietary, restricted or competition sensitive information contained in their response. The Government will not pay for the information submitted in response to this RFI. Generic capability statements will not be considered. Responses must address capabilities specific to this RFI.

2. Problem Statement

The Army's current acquisition and fielding processes for commercial off-the-shelf (COTS) hardware lack the responsiveness and flexibility required to keep pace with rapidly evolving user mission requirements and technological advancements. The rigid procedures across several Army regulations and policies fail to adapt to the dynamic nature of modern warfare and technological innovation, resulting in a significant mismatch between the Army's operational needs and the capabilities of the systems it deploys. The current processes often lead to the procurement of outdated equipment that is approaching obsolescence before it reaches the end-user, wasting resources and potentially compromising mission effectiveness. Consequently, there is an urgent need to develop a more agile and adaptive acquisition, fielding and sustainment process that can quickly respond to mission requirements and technological improvements, ensuring that the Army remains at the forefront of military capability and operational readiness.

3. Description of Need

PdM TMC provides software and computing capabilities to support Command and Control in Army operational units (brigade and higher) and associated schoolhouses and training centers. The software is a mix of customized Command and Control software, Government Off the Shelf (GOTS) software, and commercially licensed enterprise software, including virtualization infrastructure. PdM TMC also provides computing capabilities – CPU, memory, and storage – to support operations in DDIL environments. PdM TMC is challenged by time consuming and restrictive Army processes that severely impede our ability to provide the most up-to-date computing capabilities to our customers and to be responsive to changing demands. PdM TMC needs a solution to provide not only initial deployment of computing capabilities but an ongoing sustainment and refresh of those capabilities.

4. Background

4.1. Current State: PdM TMC fields operational mission command capability to the US Army. PdM TMC currently meets objectives through the fielding of multiple distinct products: warfighting applications (WFA), enterprise/common software services, virtualization infrastructure, and tactical server hardware. While fielded units own their products once they are fielded, PdM TMC remains the lifecycle manager of these products.

4.2. These products are fielded to units across the US Army, each with their own unique mission sets. Alignment of specific software and hardware to each unit's needs requires a rigorous process that results in a basis of issue (BOI) for each unit.

4.3. PdM TMC currently manages four (4) hardware families that are in use by Army units. Some hardware families have sub-variants that provide various levels of size, weight, power, and compute (SWAPC) capability:

- Battle Command Common Services v5 (BCCSv5)
- Tactical Server Infrastructure v1 (TSIv1)

- Tactical Server Infrastructure v2 (TSIv2) – Large and Small
- Tactical Server Infrastructure v3 (TSIv3) – Large, Medium, and Small

Mission Command Software Resource Requirements

PdM TMC fields software capability along with a host of other WFA developers to units with various roles. Below are the rollups of current software resource utilization at three of these roles.

Main Command Post

- 160 CPU Cores
- 708 GB Memory
- 38 TB Storage

Tactical Command Post

- 151 CPU Cores
- 640 GB Memory
- 30 TB Storage

Mobile Command Post

- 44 CPU Cores
- 168 GB Memory
- 6 TB Storage

4.4. Responsive modernization of COTS hardware is hampered by three main challenge areas:

4.4.1. First, is the timely identification and approval of hardware for fielding. The approval process is the testing and certification of the hardware (or hardware and software) IAW AR 770-3 Type Classification and Materiel Release. Materiel Release is the Army's process to prove a weapon system is Safe, Suitable, and Supportable. COTS hardware is subject to rapid Obsolescence and Diminishing Manufacturing Sources and Material Shortages (DMSMS) cycles. Each iteration of hardware can be subject to new approval processes under this regulation. However, the approval process including all applicable testing and certifications can take upwards of 9-12 months to complete almost ensuring a 3rd to a 5th of any COTS solution useful lifespan is expended before procurement even starts.

4.4.2. Second, is the Materiel Fielding process defined in AR 770-2. The documentation, planning, and prioritization activities to execute either Total Package Fielding of new systems or Modified Work Orders (MWOs) (defined in AR 750-10) is also lengthy and depends on authorizations from multiple organizations including HQDA G3/5/7, U.S. Army Forces Command (FORSCOM), Army COMPOs, and Army Materiel Command. Prioritization of fielding and training activities is defined by Army's Regionally Aligned Readiness and Modernization Model (ReARMM). The model was implemented to balance the demands of unit organization activities and fieldings, however Re-ARMM can limit when a Product Office can execute fielding and training to units by months or years.

4.4.3. Third, is the process of Transition to Sustainment (T2S) defined in AR 700-127. After a weapon system production is complete and the Product office has finished fielding and training activities to the weapon system's Basis of Issue (BOI) there is a highly structured process to transition the system from the Production to Operations & Support lifecycle phase. This includes restriction or elimination of any technical refresh of the systems and/or their sub-components, effectively freezing the system's configuration. There can also be financial restrictions post T2S that reduce how much money can be spent on hardware using the Operation and Maintenance, Army (OMA) funding available in this lifecycle phase. Significant changes required to sustain a system may trigger a new cycle of RDT&E, Materiel Release, Production and Fielding.

5. Response Format

This is an initial RFI asking for industry ideas on how to solve PdM TMCs ongoing fielding and sustainment problems. We are requesting white papers describing service approach including proposed technical solutions you would use to provide services.

White papers should be no longer than ten (10) pages.

First two (2) pages should provide your service approach. Please explain your high-level approach to providing infrastructure as a service. In your approach, please provide information pertaining to:

- Cost models you would utilize (e.g., consumption vs flat rate)
  - Consumption estimate models

  - Continuity of operations service model
- Edge hardware
  - Ownership model
  - Hardware selection
  - Service failure resolution
- Software infrastructure service model
  - Gov provided vs service provided
  - Interoperability
- Field / Theater Support
  - Garrison support
  - Operations support
- Authority to Operate (ATO)
- Refresh Cycle Planning and Triggers
- Additional Concerns
- What other areas should PdM TMC consider?

Remainder of white paper is open to additional details on your proposed solutions, implementations, and past performance.

6. Technical Capabilities

When providing details on your proposed solution, please consider the following capability needs.

- How will end users (soldiers) interact with you proposed solution (hardware and / or software)
  - Automation
  - Documentation
  - Monitoring and dashboards
  - Service and data management
- How will the soldier be able to customize or tailor their delivered solution to meet their specific business or mission needs?
  - Scalability to meet demand
  - Integration with existing unit-owned hardware
  - Integration with unit-leased cloud instances
  - Unit-level software deployments
- How will you ensure service availability in contested environments?
  - Performance
  - Disaster recovery
  - Data security
  - Regulatory compliance

- How will your solution be implemented and sustained?
  - Supplemental Training Materials (Technical Manuals, Quick Reference Guides, etc.)
  - Training Execution (Synchronous or Asynchronous)
  - Fielding / Distribution
  - Property Accountability via recognized Accountable Property System of Record (APSR)
  - Help Desk and/or Field Support
  - Maintenance, Repair, Returns and Replacement

6.1. Key requirements include high availability, robust performance, seamless integration with existing Command and Control systems, and flexible pricing models that align with our consumption patterns. Additionally, the solution should include comprehensive management tools, monitoring capabilities, and support for hybrid cloud environments to enable smooth integration with our on-premises infrastructure. Security features, such as encryption, identity management, and compliance certifications are critical to ensure data integrity and adherence to industry standards.

6.2. Vendors should demonstrate technical expertise, reliable support services, and a proven track record in delivering scalable and innovative IaaS solutions. Responses should provide solutions for all phases of real-world Army unit training and operations, including home station, exercises, training environments, and wartime deployments.

6.3. Deployment of physical hardware should present minimized SWAP as well as providing currency – hardware should be no more than one year old at any given point in time.

7. Operational Use Cases

These operational use cases are provided as a non-exhaustive list of some of the types of operational needs for which PdM TMC provides capability.

7.1. General Use Case Requirements

a. Applications and data must be continuously available for soldier use, independent of external network connectivity or hardware failure.

b. Solutions must be tailorable to individual units based on mission set and standard operating procedures.

7.2. Unit Exercise / Deployment

7.2.1. Garrison

a. Unit is in garrison and performs routine operations using warfighting applications. Data products are created and used in day-to-day operations with end-user devices.

b. Unit is assigned to an exercise that requires mobilizing to another physical location.

c. Unit prepares to mobilize

1) Unit receives any needed episodic edge hardware, potentially including end-user devices.

2) Applications and data products are available on received equipment.

3) Home station instance of applications and services remains operational.

4) Physical packaging of equipment

7.2.2. Forward Location

a. Unit travels to the exercise / deployment location with packaged equipment.

b. Unit unpackages equipment and conducts exercise.

c. Throughout operations, data products are updated and must also synchronize with home station.

d. Mission is completed.

7.2.3. Garrison

a. Unit packages equipment and transitions back to garrison.

b. Unit migrates applications and data products from received equipment if needed.

c. Unit returns episodic equipment

1) Declassification

2) Packaging and return

d. Unit continues routine home station operations.

7.3. Combat Training Center

a. Training center staff and instructors perform daily operations of developing and maintaining operational workflows and associated data products in preparation of unit exercises.

b. Training unit arrives and conducts exercise, which may be destructive to the training environment.

c. Exercise concludes.

d. Training center staff reset the training environment in preparation for the next unit exercise.

7.4. Training and Simulation Environments

a. Host realistic training and simulation environments for commanders and staff.

b. Enable virtual wargaming, mission rehearsal, and cyber defense exercises.

c. Leverage cloud-based infrastructure instead of physical training facilities where appropriate.

7.5. Remote Management & Automated Infrastructure Scaling

a. Unit IT teams can remotely provision, monitor, and scale infrastructure as mission demands fluctuate.

b. Reduce the logistical burden of transporting and maintaining physical hardware in deployed environments.

Attachments/Links

No attachments or links have been added to this opportunity.

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